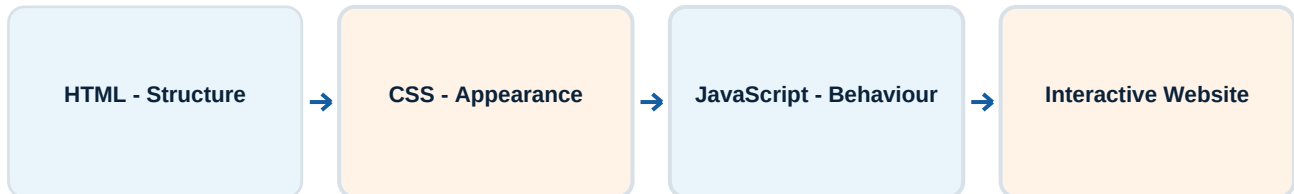


JavaScript Fundamentals

Class 01 - From a static webpage to an interactive webpage

What is JavaScript?

JavaScript is a programming language used to make websites **interactive and dynamic**. HTML gives a webpage its structure, CSS controls how it looks, and JavaScript controls what happens when the user interacts with it.



A simple way to remember it

HTML	Creates elements such as headings, paragraphs, forms and buttons.
CSS	Adds layout, spacing, colors, fonts and responsive design.
JavaScript	Adds actions such as clicks, alerts, content changes and dynamic updates.

Example

```
<button>Click Me</button>

button {
  background-color: blue;
  color: white;
}

alert("Button clicked!");
```

What JavaScript can add to a webpage

- Change text and styles
- Show alerts
- Respond to clicks
- Validate forms
- Show/hide content
- Update data
- Call APIs
- Build interactive applications

Key idea: HTML + CSS can create a complete static page. JavaScript adds behaviour and interaction on top of it.

Setting Up JavaScript in VS Code

You can run JavaScript in a browser immediately. Node.js is useful when you want to run JavaScript from the terminal or outside the browser.

Install Node.js using Winget

Open Windows Command Prompt and run:

```
winget install OpenJS.NodeJS.LTS
```

After installation, close and reopen the terminal. Verify the installation:

```
node -v  
npm -v
```

The exact version numbers may be different on each system. That is normal.

If Winget does not work

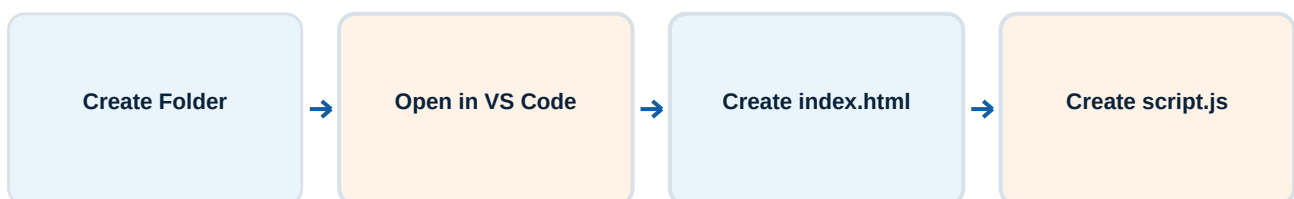
Open the official Node.js website: **nodejs.org**. Download the **LTS** version and install it using the normal Windows installer. Then verify again with `node -v` and `npm -v`.

Why Node.js?

JavaScript already runs inside Chrome, Edge and other browsers. Node.js allows the same language to run outside the browser, including from the VS Code terminal.

```
// script.js  
console.log("Hello Students");  
  
# Run from terminal  
node script.js
```

VS Code project setup



Important: Node.js is not a compiler for JavaScript. It is a JavaScript runtime environment.

Your First JavaScript in Browser DevTools

The browser Console is one of the easiest places to test JavaScript before connecting it to a project.

Open DevTools



You can also press **F12** in most browsers.

Hello World

```
console.log("Hello World");
```

Expected output:

```
Hello World
```

Try simple values

```
console.log("Welcome Students");  
console.log(10 + 20);  
  
let studentName = "Gowri";  
console.log(studentName);
```

Show an alert

```
alert("Welcome to JavaScript!");
```

Why use console.log()?

Developers use `console.log()` to check values, test code, understand execution flow and debug problems.

Class idea: Keep DevTools open while experimenting. The Console gives immediate feedback, which is perfect for beginners.

Connecting JavaScript with HTML

Now we connect an external JavaScript file to the webpage and use JavaScript to access HTML elements.

Project files

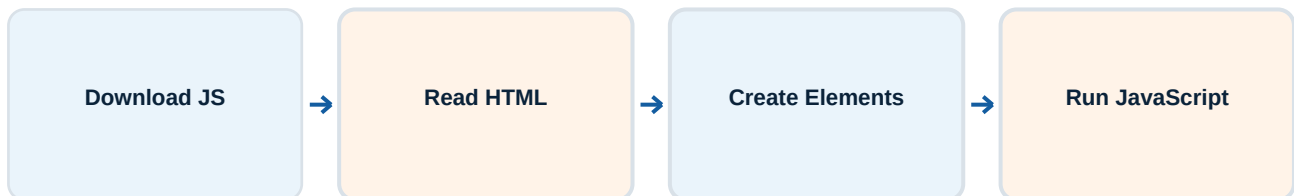
```
project/  
|-- index.html  
|-- script.js
```

index.html

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>JavaScript Class</title>  
  <script src="script.js" defer></script>  
</head>  
<body>  
  <h1 id="title">Welcome Students</h1>  
  <p id="message">We are learning JavaScript.</p>  
  <button id="changeButton">Click Me</button>  
</body>  
</html>
```

Why use defer?

defer lets the browser continue reading and creating the HTML page before the JavaScript executes. This prevents JavaScript from trying to access an element before that element exists.



Understanding document

document represents the current HTML page inside JavaScript. We use it to find and work with elements.

```
document.getElementById("title");
```

The line above means: **look inside the current document and find the element whose id is title.**

DOM Manipulation and Button Behaviour

Once JavaScript finds an HTML element, it can change its text, style and behaviour.

Change text using textContent

```
const title = document.getElementById("title");  
  
title.textContent = "Welcome to JavaScript";
```

textContent reads or changes the text inside an HTML element.

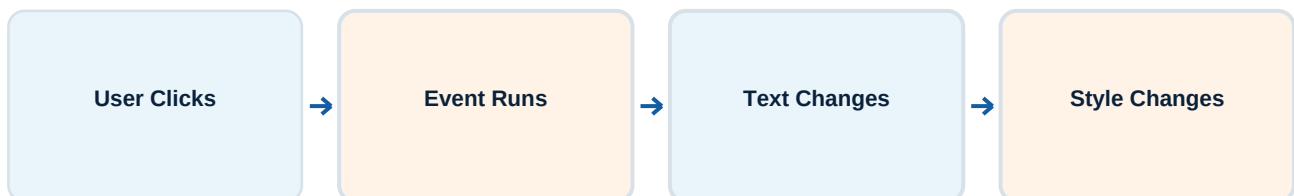
Change styles

```
title.style.color = "blue";  
title.style.backgroundColor = "yellow";
```

CSS uses background-color, while JavaScript uses backgroundColor.

Final button example

```
const title = document.getElementById("title");  
const message = document.getElementById("message");  
const button = document.getElementById("changeButton");  
  
button.addEventListener("click", function () {  
    alert("Button clicked!");  
  
    title.textContent = "Welcome to JavaScript!";  
    title.style.color = "blue";  
  
    message.textContent = "JavaScript changed this text.";  
  
    button.textContent = "Clicked";  
    button.style.backgroundColor = "green";  
    button.style.color = "white";  
});
```



Before -> After

Before clicking	After clicking
Welcome Students We are learning JavaScript. [Click Me]	Welcome to JavaScript! JavaScript changed this text. [Clicked]

Practice: Change the heading color, change the alert message, make the button say "Done", and change the paragraph text to your own sentence.